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Project Title:	Investigation of Neural Strategies of Visual Search		
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	Human performance is easily impaired in a variety of aerospace tasks – such as remotely docking a spacecraft, or flying an aircraft in poor visibility, or finding and tracking a particular target airplane on a crowded air-traffic control display. These impairments occur because aerospace environments can degrade or remove the sensory cues used to drive and coordinate voluntary actions, and display systems may not be optimally matched to the human user. Fortunately, much progress has been made recently in the field of neuroscience toward understanding the physiological mechanisms responsible for voluntary motor behaviors. In particular, we are beginning to understand how the brain accomplishes visual search – the process that allows us to quickly respond to important items in our surroundings, even in the presence of other irrelevant stimuli. Understanding how individual neurons participate in this process is critical for identifying the factors that limit the speed and accuracy of visual search. Knowledge of these neural factors could be used to design visual displays that conform to the temporal and spatial constraints imposed by human neurobiology, thereby directly		

	improving human performance in man-machine interfaces and virtual environments. Arguably the most important class of movements made by primates during visual search are eye movements, because control of the position of the eyes determines which visual information is sampled and for how long. Decades of anatomical and physiological research have provided a detailed outline of the brain pathways that are responsible for the control of eye movements in primates. These pathways involve a network of areas including visual areas of the cerebral cortex (which provide information about the content of the visual scene), motor regions of the cerebellum and brainstem (which form the final motor commands), and several regions that lie in-between (and which possess a mixture of sensory and motor properties). It is these latter, intermediate, brain regions that are most likely responsible for forming the decision about where the eyes should move next during visual search, because they lie at the interface between the visual areas that provide evidence about the location of stimuli and the motor regions that execute the movements. One of these structures, the superior colliculus (SC), has long been known to play a pivotal role in the control of saccadic eye movements. However, most studies of the SC have focused on its contribution to the motor execution of saccades. Because of the importance of the SC in the decision processes that precede saccade generation, examination of this structure during visual search is likely to reveal fundamental aspects of the visual neural computations involved in identifying, localizing, and selecting visual targets. The main goal of this project would be to measure how neurons in the superior colliculus (SC) change their activity during a visual search task. Specifically, we propose to measure how the activity of these neurons is altered by the discriminability of visual targets and test whether these changes can predict the changes in the subjects' performance. Understanding how the information encoded by these neurons constrains overall search performance will foster the development of more predictive models of human performance and ultimately lead to the design of better aerospace displays and interfaces.
Task Description:	Goal: To develop a computational model of decision-making, involving multiple sources, agents and outputs, based on neurophysiological data. The phenomenon: Visual search is a basic form of decision-making that is performed by humans and non-human primates several times a second (e.g., as you read this text). The brain pathways that accomplish this behavior have been identified, but the processing steps and computations that are involved in supporting this phenomenon are not known. The experimental approach: By recording from single neurons in the brains of non-human primates as they perform visual search, we have a window into the internal processing steps that support visual search and how these change over time on a millisecond time scale – data that cannot be obtained from behavioral experiments with humans. In addition, we conduct parallel behavioral experiments with humans to confirm the applicability of this approach. The hypothesis: By recording the activity of neurons during visual search, we can identify the biological computations that underlie performance in visual-motor tasks and decision-making. Countermeasure readiness level: 2 at start of project, 3 now. The original hypothesis has now been validated and we are in the process of understanding in detail the neural processes steps that support neurally-mediated visual search. Specifically, we are testing how different sources of information are combined and read out by neurons during this basic decision task. Expected countermeasure readiness level at end of project: 4 to 5. Once we understand the dynamic regulation of top-down and bottom-up factors, we will have identified the key features that are relevant for constructing biologically inspired models of visual search. We will then formulate how these features should be incorporated into computer-based models of decision-making (level 4), and then prove the feasibility and efficacy of this approach by constructing a model of visual search based on these biological features (level 5). After this initial demonstration, the concept may be applied to other processes that would benefit from automated decision-making (levels 5 – 6). The results from these experiments provide new insights into the biological mechanisms underlying visual search behavior in humans and other primates. Although the immediate goals of the project were to expand our basic biological knowledge, the results are also be applicable to the stated goals of NASA's Space Human Factors, Intelligent Systems, and Aerospace Operation Systems Programs. Biological systems are capable of accomplishing tasks that have proved extremely difficult to duplicate using artificial intelligence and robotic systems – especially the job of detecting, discriminating and responding to stimuli quickly and accurately, even in the presence of background noise. Our results have provided new data about the algorithms and information processing streams used by biological systems to accomplish such tasks, and may therefore aid in the development of robust computational models of human visuomotor performance and in the design of the autonomous systems for future aerospace applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of these neural factors could be used to design visual displays that conform to the temporal and spatial constraints imposed by human neurobiology, thereby directly improving human performance in man-machine interfaces and virtual environments.
Task Progress:	Most of our effort during this review period was devoted to training subjects on a new visual search task in which the instructions change unpredictably. The significance of this pproach is that visual search is often performed under changing task demands, but the neural mechanisms responsible for performance monitoring are largely unknown. By testing the two major brain regions involved in visual search under these conditions, we expect to identify key properties of the underlying neuronal mechanisms. Our working hypothesis is that changes in visual search caused by "bottom-up" changes in visual signals will be largely mediated by the superior colliculus, whereas changes in visual search caused by "top-down" changes in cognitive instructions will be largely mediated by the frontal eye fields in the cerebral cortex. To test this idea, we have trained a monkey to perform a task that requires him to change the goal of their movement based on either a visual or symbolic instruction. Using an analysis procedure based on signal detection theory, we will identify when neurons discriminate the two types of instructions. Our prediction is that SC neurons will discriminate visual instructions sooner than FEF neurons, but that FEF neurons will discriminate symbolic instructions sooner than SC neurons. We have also developed a novel task to examine how knowledge of target probability affects visual search. In these new studies, we have compared the ability of humans to learn a Markov process, in which target probabilities depend on the

	correct answer of the previous trial, to their ability to learn an independent and identically distributed (IID) process, in which target probabilities do not depend on history. Our results to date show that subjects can benefit from a Markovian probability structure when selecting targets, indicating that the process guiding saccade selection includes the ability to incorporate patterns in assigning target probabilities. We presented this work at the Society for Neuroscience meeting in Washington, D.C.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
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